SUPPORTING ADULT GUIDANCE

The tangent ratio

Introduction to Trigonometry. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner meets the first named trigonometric ratio: the **tangent**, defined as **tan θ = opposite ÷ adjacent**. The emphasis is that this ratio is **constant for a given angle** (so it is a property of the angle, not the size), that it **increases as the angle increases**, and that it can be **used to compare steepness**. The hypotenuse is deliberately not involved.

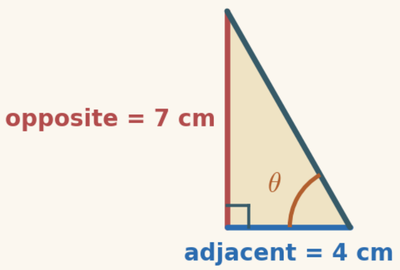
Driving Question: *How does opposite ÷ adjacent capture an angle?*

The mathematics, in plain terms

This lesson names the first of the three trigonometric ratios. For a chosen acute angle θ in a right-angled triangle, the **tangent** is **tan θ = opposite ÷ adjacent** — the side opposite θ divided by the side adjacent to θ. Two ideas from earlier lessons make this meaningful. First, from Lesson 1, this ratio is the **same for every triangle with that angle**, so tan θ is a single number that **belongs to the angle**: a 3-4 triangle and a 6-8 triangle both give 0.75. Second, from Lesson 2, 'opposite' and 'adjacent' are defined **relative to the angle**, which is why the tangent is tied to a particular angle. The new perception is that as the **angle grows, the tangent grows**: a steeper slope has a bigger tangent. This is exactly the idea of **gradient** from the Straight Lines unit — rise over run — now attached to an angle, and it is why the tangent is such a natural measure of steepness. The tangent deliberately **does not use the hypotenuse**, which keeps this first ratio simple. The two errors to watch for are **using the hypotenuse** in place of the adjacent, and **inverting** the ratio to adjacent ÷ opposite; both are headed off directly in the lesson. Sine and cosine follow next, organised as SOH CAH TOA, so getting the **order** of the tangent ratio secure — opposite on top — matters now.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

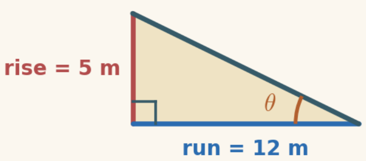


Example 1 — finding the tangent

The tangent of the marked angle is the opposite divided by the adjacent. Here the opposite is 7 cm and the adjacent is 4 cm.

1. the tangent uses opposite and adjacent	$\tan\theta = \dfrac{\text{opposite}}{\text{adjacent}}$
2. put in the numbers	$\tan\theta = \dfrac{7}{4}$
3. work it out	$\tan\theta = 1.75$

tan θ = 1.75



Example 2 — the tangent of a ramp

A ramp rises 5 m over a run of 12 m. Its rise is the opposite and its run is the adjacent, so its tangent is rise ÷ run.

1. the tangent uses opposite (rise) and adjacent (run)	$\tan\theta = \dfrac{\text{rise}}{\text{run}}$
2. put in the numbers	$\tan\theta = \dfrac{5}{12}$
3. work it out	$\tan\theta \approx 0.42$

tan θ ≈ 0.42

A number that measures an angle

The heart of this lesson is that a single ratio can **stand in for a whole angle**. It is worth letting two ideas settle. First, **constancy**: tan θ is the same whatever the size of the triangle, because the angle fixes the ratio (Lesson 1). This is what makes it worth naming — if it changed with size it would be useless. Second, **monotonicity**: as the angle increases the tangent increases, so a bigger tangent always means a steeper angle. Grounding both in the learner's own work — computing opposite ÷ adjacent for two sizes and getting the same answer, then watching the tangent climb as the slider angle grows — makes the idea convincing rather than asserted. The link to **gradient** (rise over run) from Straight Lines is worth drawing explicitly: the tangent of the angle a line makes with the horizontal is exactly its gradient. Keeping the ratio the right way up (opposite ÷ adjacent) and off the hypotenuse is the precision that the next lessons, where sine and cosine join in, will depend on.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The ratio with a name: opposite ÷ adjacent is the same for every size of the same-angle triangle — that number is tan θ.	~10 min
Movement	Tangent Explorer: drag the angle; the opposite side and tan θ grow with it while the adjacent stays fixed.	~14 min
Reflection	Is this tan θ?: judge claimed tangents, catching the hypotenuse and upside-down errors.	~10 min
Creativity	Rank by steepness: work out tan θ for three slopes and order them from gentlest to steepest.	~6 min
Rest	A pause after seeing a whole angle captured by a single number.	~3 min
Nutrition	Mode B (intellectual): what the learning feeds — letting one well-chosen number stand for something larger.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using the hypotenuse instead of the adjacent

Repair: "tan θ uses the opposite and the adjacent only — not the hypotenuse. Which side is adjacent to θ?"

Inverting the ratio to adjacent ÷ opposite

Repair: "Which side goes on top? tan θ = opposite ÷ adjacent, opposite first."

Thinking tan θ changes when the triangle is enlarged

Repair: "Work out opposite ÷ adjacent for the small one and the big one. Same angle — same tangent."

Thinking a bigger triangle has a bigger tangent

Repair: "Size does not change tan θ. Only a bigger angle does — that is what steeper means."

Reading tan θ as an angle in degrees

Repair: "tan θ is a ratio, a plain number. It measures the angle, but it is not the angle in degrees."

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: naming the opposite and adjacent sides relative to a marked angle (Lesson 2), the constant side ratio for a fixed angle (Lesson 1), and a ratio written as a division (a ÷ b). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the definition and the two ideas of constancy and steepness with the learner:

- “Which side is opposite θ, and which is adjacent? So what is tan θ?”
- “We enlarged the triangle — did tan θ change? Why not?”
- “The angle got bigger — what happened to tan θ?”
- “Two slopes: one has tan θ = 0.4, one has tan θ = 0.9. Which is steeper?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why is it opposite ÷ adjacent and not the other way round?”
That is simply the definition of the tangent, and keeping the order fixed matters: opposite ÷ adjacent increases as the angle increases, which is what makes tan θ a natural measure of steepness. Getting the order secure now prevents errors when sine and cosine arrive.

“Why does the hypotenuse not appear?”
The tangent compares only the two sides around the right angle — the opposite and the adjacent. Leaving the hypotenuse out keeps this first ratio simple; the hypotenuse joins in with sine and cosine in the next lesson.

“How does this connect to gradient from Straight Lines?”
Very directly: the gradient of a line is rise over run, and the tangent of the angle the line makes with the horizontal is opposite over adjacent — the same thing. A steeper line has a bigger gradient and a bigger tangent.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.

Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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